



per cent of the emission reductions will, in all probability, need to come from SAF.

SAF represents a broad category of fuels derived from non-fossil sources, including advanced biofuels and electro-fuels (e-fuels). However, the real challenges lie in logistics and costing. “The problem globally, and for India, is ensuring that SAF actually gets produced and the logistics of bringing it from the site of production to the bowser that goes to refill the aircraft,” says Scindia. Similarly, depending on whether it is produced from recycled waste or synthetic fuel, it can be two to five times costlier than standard jet fuel, according to IATA estimates. And this is where Minister Puri senses India’s potential to become a surplus and cost-effective producer of SAF. “India has feedstock for potential production of 19-24 million tonnes (MT) of SAF per year, whereas the estimated maximum requirement of SAF in India, considering 50 per cent blend, is around 8-10 MT per year by 2030,” he declares. The biofuel powering the AirAsia India flight, for instance, was developed using alcohol-to-jet technology, using sugarcane molasses as feedstock.

FROM AIRLINES TO AIRPORTS

The environmental, social and governance (ESG) strategy of IndiGo—India’s largest carrier by fleet and market size—is premised on balancing the ambition and practical limitations of the aviation sector. Since 2015, the low-cost carrier has been working towards reducing emission intensity by 18 per cent by 2023. Highlighting the role of technology, IndiGo CEO Pieter Elbers says, “Today, it may look like there has hardly been any change in aviation technology. However, if you see how an Airbus 321neo compares in fuel consumption to a plane 15 years ago, hardly any industries have created such a step up in efficiency.” Consequently, the airline has been consistently adding more fuel-efficient Airbus A320ceo and A321neo aircraft to acquire one of the youngest fleets globally. Other measures include single-engine taxiing, reduction in hold fuel to lower aircraft weight and route savings.

Air India is also making a concerted effort at sustainability as a Tata group firm. “It adds to the impetus to do more when it comes to environmental responsibility. Making sure that we have a young and modern fleet; that we have all of those things that save weight on an aircraft;

PHOTO BY YASIR IQBAL



“THE PROBLEM GLOBALLY, AND FOR INDIA, IS ENSURING THAT SAF ACTUALLY GETS PRODUCED AND THE LOGISTICS OF BRINGING IT FROM THE SITE OF PRODUCTION TO THE BOWSER THAT GOES TO REFILL THE AIRCRAFT”

JYOTIRADITYA SCINDIA
MINISTER OF CIVIL AVIATION